

A Mechanistic Study of Transformers Training Dynamics

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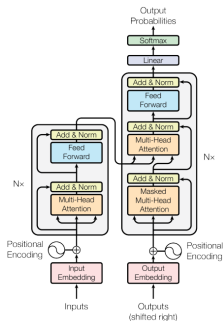
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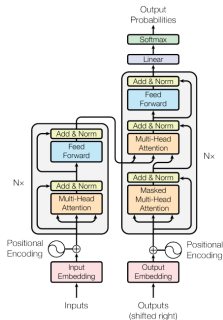
Most foundation models are based on the transformer architecture.



(Vaswani et al., 2017)



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- Diverse training data
- Large-scale pretraining
- (Very) large models



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- ✓ Better understand models to prevent harmful behaviors



- ✓ Reverse-engineer the computations performed by transformers
- ✓ Better understand models to prevent harmful behaviors
- ✓ Controlled settings and simplified models

A Mathematical Framework for Transformer Circuits

AUTHORS

Nelson Elhage^{*,†}, Neel Nanda^{*}, Catherine Olsson^{*}, Tom Henighan[†], Nicholas Joseph[†], Ben Mann[†], Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, Chris Olah[‡]

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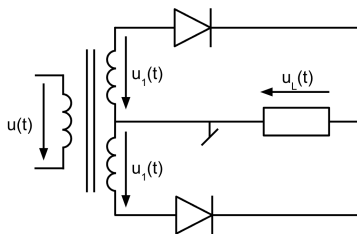
Anthropic

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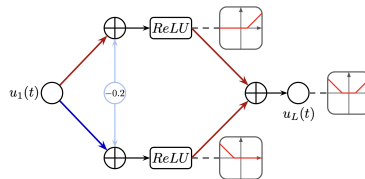
Dec 22, 2021



- ✓ Human brain $\rightarrow$ complex electrical circuit of neurons,
- ✓ Flow of information controlled by rectifiers,
- ✓ In deep learning, rectifiers are everywhere: ReLU activations.



(a) Electrical circuit.



(b) Neural network implementation.



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Chris Olah et al. (Anthropic) studied the in-context learning phenomenon from the perspective of induction head circuits.

In-context Learning and Induction Heads

AUTHORS

Catherine Olsson*, Nelson Elhage*, Neel Nanda*, Nicholas Joseph†, Nova DasSarma*, Tom Henighan†, Ben Mann†, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, Chris Olah†

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A Famous Example: Induction Heads

Alberto Bietti et al. (2023) studied memorization and generalization in transformers using induction heads.

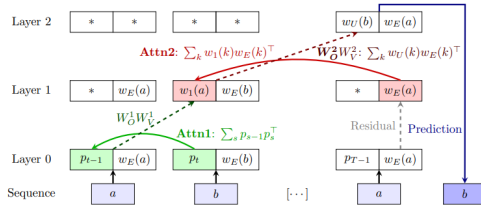
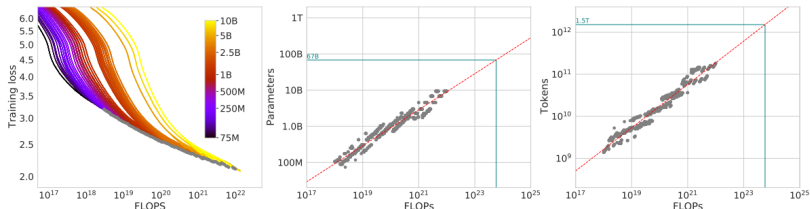


Figure 1: **Induction head mechanism.** Induction heads are a two-layer mechanism that can predict b from a context $[\dots, a, b, \dots, a]$. The first layer is a *previous token head*, which attends to the previous token based on positional embeddings ($p_t \rightarrow p_{t-1}$) and copies it after a remapping ($w_E(a) \rightarrow w_1(a) := W_O^1 W_V^1 w_E(a)$). The second layer is the *induction head*, which attends based on the output of the previous token head ($w_E(a) \rightarrow w_1(a)$) and outputs the attended token, remapped to output embeddings ($w_E(b) \rightarrow w_U(b)$). Boxes in the diagram represent different embeddings in superposition on each token's residual stream (we omit some irrelevant ones for clarity, e.g., positional embeddings in upper layers), and attention and output associations are shown with the associative memory viewpoint presented in Section 4.



Models' performance will continue to improve as the size of the model and its training set increase (Kaplan et al., 2022; Hoffman et al., 2022).





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- ✓ Significant resources dedicated to training very large models,
- ✓ Better understanding training dynamics → minimize this cost.



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- ✓ Input: sequences of N tokens in the ring of integers modulo p ,
- ✓ Targets: sum of the first k terms modulo p .
- ✓ Input length N , vocabulary size p , sparsity index $k \in [N]$,
- ✓ In practice, we take $p \in \{2, 3\}$, $N = 12$, and $k = 5$.

$$\begin{aligned} f^* : \quad & (\mathbb{F}_p)^N && \rightarrow && \mathbb{F}_p \\ & x = (x_1, \dots, x_N) && \mapsto && \sum_{t \in [k]} x_t. \end{aligned} \quad (\text{SMA})$$

The model consists of a single transformer block with cross-attention.



Outputs y has suffix and permutation invariances w.r.t. inputs x .

Clustering head solves the SMA problem as follows:

- ✓ Embedding ensures invariance;
- ✓ Sequence embeddings are clustered;
- ✓ The FFN scatters into decision regions.

Theorem. Let θ be the parameters of a transformer model that implements a clustering head. Then, θ is a stationary point of the loss, that is $\nabla \mathcal{L}(\theta) = 0$.



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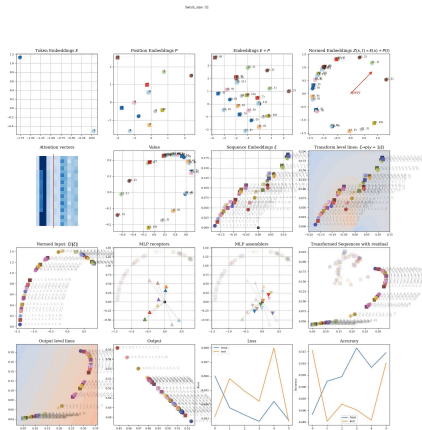


Tokens can be seen as a system of 2D interacting particles.

We monitor during training:

- ✓ Position & token embeddings;
- ✓ Attention map & value transform;
- ✓ Sequence embeddings;
- ✓ FFN probability mapping;
- ✓ Loss and accuracy.

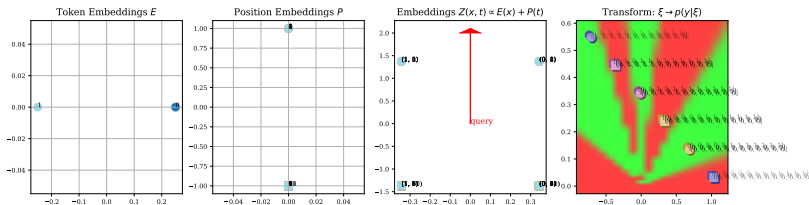
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Clustering head implementation



- ✓ Spurious (squares) and non-spurious (circles) tokens collapsed;
- ✓ Query (red arrow) points towards the non-spurious embedding;
- ✓ Attention mechanism focuses solely on non-spurious tokens.
- ✓ Clustered embeddings, respecting the invariance;
- ✓ The FFN maps each cluster to its class.

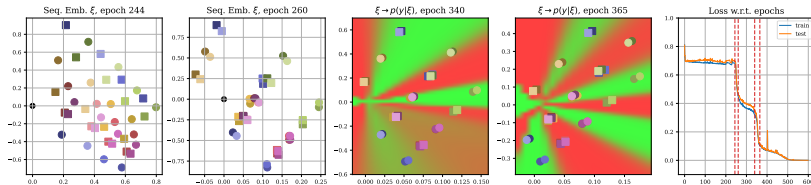


Practical clustering heads implementations are variations of the idealized circuit above.

Two-stage learning



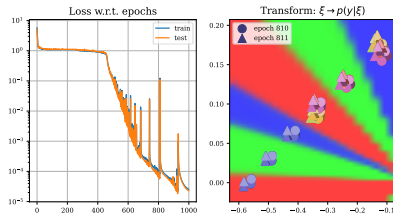
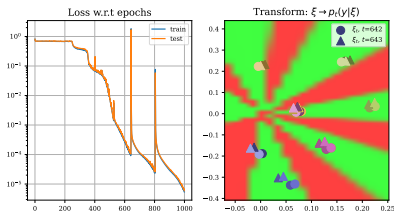
- ✓ First phase: unstructured sequence embeddings;
- ✓ Structure in embeddings emerges after the first loss drop;
- ✓ Second phase: FFN starts to classify clusters;
- ✓ The second loss drop occurs as the FFN fits the embeddings.





Loss spikes can emerge because of:

- High-curvature normalization;
- High-curvature feedforward;
- Loss spikes and gradient norms occur in tandem.





- ✗ Loss spikes occur in large-scale pretraining;
- ✗ It can lead to loss divergence;
- ✓ Controlling embedding norms can help → QK-Norm (Deghani et al., 2023);
- ✓ Controlling gradient norms can help → MuonClip used for Kimi K2 pretraining;
- ✓ New family of second-order optimizers (Muon, Shampoo, AdaFisher) can help control gradient variations.



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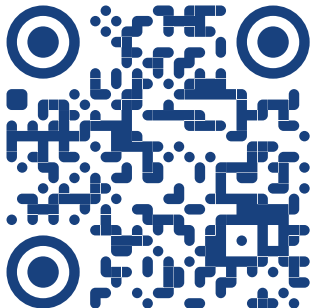


- ✓ Controlled interpretable setting;
- ✓ Visual monitoring of tokens;
- ✓ Circuits emerge during learning;
- ✓ Training occurs in two stages;
- ✓ Occurrence of loss spikes.

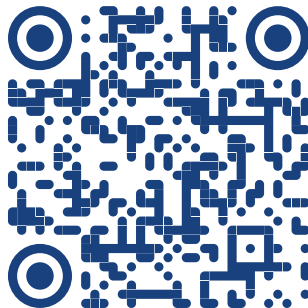
Our sandbox can be used at small scale to build intuition towards more practical settings.



Check out the paper along with the code to learn more!



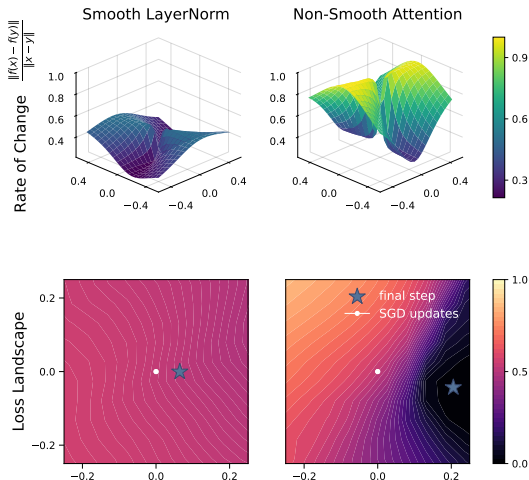
paper



code



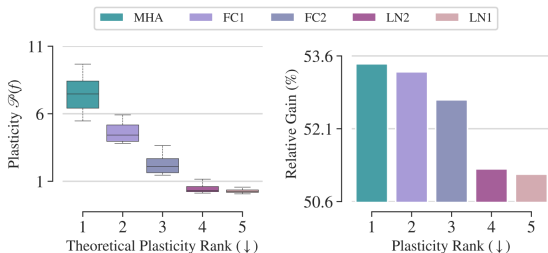
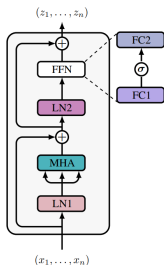
ICML 2026: ViT Benefits from Non-Smooth Components



[paper](#) - [code](#)



Finetuning high-plasticity components (non-smooth) yields larger performance gains than the stiff ones.



[paper](#) - [code](#)

Thanks for your attention!